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GenAI Adoption Increases the Density of Knowledge and Collaboration Networks: Evidence from a Field Experiment

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While generative AI (GenAI) tools can boost individual productivity, their effects on intra-organizational networks for collaboration and knowledge sharing remain largely unknown. We theorize that adopting GenAI reshapes employees' interaction patterns, because it can act as a self-automator (freeing up time for collaboration) and as a knowledge catalyst (increasing the value of individuals as sources of knowledge), thereby rewiring interactions. We further propose role-based differences in these effects across specialists (deep domain experts) and generalists (broad integrators). We test these ideas in a randomized field experiment with 316 employees at a European technology services firm, randomly assigning members of 42 teams to either use a GenAI assistant customized with organization-specific knowledge (treatment) or work as usual without it (control). After three months, employees with the GenAI tool became significantly more central in their collaboration and knowledge-sharing networks than the control group, reflecting an expanded set of connections and more frequent knowledge exchange. Specialists experienced greater increases in knowledge centrality than generalists, while generalists gained the most in terms of productivity, measured as number of projects executed during the time window. These findings provide novel causal evidence that GenAI adoption can rewire organizational social structures towards greater connectivity and confirms prior research that it can augment performance differentially.

Key words: Generative AI; Organizational Networks; Field Experiment

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Introduction

Research on generative AI (GenAI) in organizations has accelerated, with early studies centering on human-AI interaction and assessing whether AI tools improve an individual's output (e.g., faster drafting, analyzing or coding). Typical findings indicate sizeable but heterogeneous gains in individual productivity (Noy and Zhang 2023, Boussioux et al. 2024, Chen and Chan 2024, Hui et al. 2024, Jia et al. 2024). Yet because organizations accomplish most of what they do through patterned collaboration rather than atomistic effort, observed productivity gains from GenAI may partly arise from changes in how employees coordinate and exchange knowledge, not only through faster individual task execution (Kilduff and Brass 2010, Puranam 2018). Existing research has not yet examined how GenAI reshapes collaboration (co-production and joint task execution) and knowledge-sharing (advice seeking and expertise access) structures, nor the mechanisms through which such changes occur. As a result, we know relatively little about the effects of GenAI adoption on patterns of collaboration and their implications for human-human interaction in organizations.

We focus on two important and distinct channels through which technology can alter the returns on collaborative effort: by changing the costs of coordination and by changing the benefits of seeking knowledge from others. These two channels correspond to different types of networks within organizations, namely collaboration networks and knowledge networks. Although these networks may overlap, they are conceptually distinct. Collaboration networks reflect co-production relationships (working closely to achieve objectives), whereas knowledge networks reflect consultation relationships (reaching out for expert advice). GenAI tools, particularly those that have been customized to the local context, e.g., through fine tuning or retrieval-augmented generation (RAG), can alter an individual's centrality in the collaboration

network by altering how easy it is for them to work with others and in the knowledge-sharing network by altering their value as a source of knowledge.

Our baseline prediction is that GenAI adoption reshapes both collaboration and knowledge-sharing networks within organizations by increasing their density. We refine this prediction by distinguishing how these effects differ for specialists, who possess deep expertise in a narrow domain (Becker 1962), and generalists, who are adequately competent in multiple domains (Teodoridis 2018, Fahrenkopf et al. 2020, Raveendran et al. 2022, Koehler et al. 2024). We propose that the costs of collaborative effort after AI adoption should fall more for specialists than for generalists, because the background knowledge gaps between specialists and their interlocutors is generally higher. In addition, the knowledge of specialists is harder to substitute than that of generalists, making the benefits of delivering their knowledge using GenAI relatively more valuable than for generalists. Thus, we should expect specialists to gain in both collaboration and knowledge centrality relative to generalists, even if both were given the same GenAI tools.

We test our predictions in a randomized field experiment at a technology services firm in Central Europe. We study 316 employees across 42 teams, randomly assigning teams to receive access to a “grounded” GenAI tool tailored to the firm’s knowledge base (treatment) or to work as usual without access to such a tool (control). Some of these employees were generalists (sales team members), whereas others were specialists (technical support). We measure outcomes over a two-week period prior to the treatment as well as three months after. Primary outcomes we measured include collaboration and knowledge-sharing interactions, the number of projects handled, and satisfaction with knowledge access. These outcomes are self-reported by employees.

The results are broadly consistent with our theoretical predictions, with some nuances. Relative to the control group, members of the treatment group exhibit an increase in the number of incoming and outgoing collaboration and knowledge-sharing ties: Gen AI adoption increases, rather than decreases the density of interactions in the organization. Productivity and satisfaction also rise in the treatment group. Further, specialists gain more than generalists in terms of centrality in the knowledge sharing network, whereas generalists gain more than specialists in terms of projects completed during our observation window. Taken together, these findings provide causal evidence that GenAI tools do not merely alter individual productivity; they can rewire human-human interaction in ways that increase user's overall density, and individual centrality in social networks, though they do so unevenly across roles. Our study thus shifts the focus from individual performance after AI adoption to AI's effects on network-level changes alongside work outcomes (Choudhary et al. 2023, Bankins et al. 2024, Hillebrand et al. 2025).

Theoretical Background and Hypotheses

Research on human-AI systems has been largely centered on individual work outcomes, framed by the automation-augmentation distinction (Raisch and Krakowski 2021, Hillebrand et al. 2025). Automation delegates tasks to AI, substituting for human effort; augmentation retains human involvement in the task, enhancing outputs for the same level of effort. Empirical studies report sizeable but heterogeneous individual gains from adopting AI tools: GenAI improves speed and quality on writing and problem-solving tasks (Noy and Zhang 2023, Boussioux et al. 2024), with benefits moderated by expertise, task type, and interface design (Allen and Choudhury 2022, Chen and Chan 2024, Hui et al. 2024, Jia et al. 2024). For instance, in a study of software developers, those with relatively lower ability realized larger benefits from an AI coding assistant, using it to concentrate more on core programming work than their higher-ability

peers did (Hoffmann et al. 2025). Existing research also emphasizes that automation and augmentation effects can co-exist (e.g., task-level automation alongside role-level augmentation), yielding nuanced patterns rather than a simple either/or dichotomy (Tang et al. 2022).

This literature establishes that GenAI can raise individual throughput and quality, particularly by offloading routine work and scaffolding complex tasks. It also documents boundary conditions (e.g. personality moderators) that shape who benefits and by how much (Tang et al. 2022, Chen and Chan 2024). In sum, existing work provides solid foundations for understanding how an individual’s adoption of AI can alter their productivity. GenAI systems can automate routine retrieval, drafting and analysis, thus improving the quality of an individual’s work by assisting with ideation, conducting checks, etc. This “co-pilot” effect can improve the quality and speed of an individual’s work.

The Effects of Grounded GenAI on Collaboration and Knowledge Centrality

This first wave of research on human AI-systems largely abstracts away from the fact that organizations accomplish work through patterns of interaction (i.e. who talks to whom, about what, and when)—not through solitary effort alone (Kilduff and Brass 2010, Puranam 2018). If GenAI changes not only the returns on solo efforts by acting as a “co-pilot,” but also on collaboration and knowledge-sharing efforts, we should expect that the pattern of interaction in an organization is itself endogenous to GenAI adoption. This network-centric view has been under-researched relative to the focus on individual productivity gains.

Grounded GenAI provides a new possibility to understand these dynamics. Unlike general-purpose GenAI tools like ChatGPT, grounded GenAI systems are embedded in firm- and role-specific knowledge; they therefore alter not only task execution but also the local economics

of exchange among coworkers in terms of the worth of their knowledge and how easy it is to exchange it (Agrawal et al. 2022, Maher et al. 2023, Zheng et al. 2023, Kemp 2024).

By automating parts of individual tasks (e.g., retrieval, summarization, or drafting), GenAI can free time (a “self-automation” effect) that employees can reallocate toward more interdependent tasks that require coordination, thereby increasing collaboration even if the tool itself is not used for collaboration. In knowledge networks, GenAI adoption can raise an individual’s value as a source by acting as a “knowledge catalyst”. Grounded, context-aware retrieval through an AI model allows people to synthesize their own know-how with organizational knowledge, yielding more tailored answers and attracting more inbound requests, possibly for more complex answers. The self-automation effect also leaves more time to act as sources of knowledge for others.

To be sure, there are countervailing forces to consider that might cause GenAI adoption to *lower* connectivity in both collaboration and knowledge networks. First, a “modularization” effect might weaken or even overturn the positive effect of Gen AI adoption on connectivity in the collaboration network. If GenAI absorbs parts of the coordination function (e.g., harmonizing multiple contributions into a single artifact), some human-human exchanges, especially routine, low-novelty ones, can disappear (Sanchez and Mahoney 1996, Baldwin and Clark 2000, Kretschmer and Puranam 2008, Puranam et al. 2009). Second, to the extent that an AI tool can answer queries nearly as well as a person, that person may be bypassed on those queries (an “expertise substitution” effect), lowering their importance in the knowledge network.

At the current stage of development and deployment of GenAI, however, we expect the positive effects on connectivity to dominate. Even when GenAI drafts or integrates knowledge work, organizations typically require human involvement for sense checking and accountability.

Incentives tied to quality, compliance, and client outcomes favor validated answers over unvetted automation, channeling traffic toward knowledgeable others. The social acceptability of AI generated outputs, unfiltered through human oversight, can be far lower than their technical acceptability as judged in a blind test (also see Dietvorst et al, 2015). In short, at the current deployment frontier, we expect the self-automation and knowledge catalyst effects to dominate the modularization and knowledge substitution effects in aggregate, increasing observed connectivity. We therefore test the following hypothesis:

Hypothesis 1. Adoption of grounded GenAI increases connectivity, (i.e. the number of ties) in both collaboration networks and knowledge networks for the adopters.

While in aggregate we expect the self-automation and knowledge catalyst effects of GenAI adoption to lead to an increase in connectivity, these effects are likely to be heterogenous in their strength across individuals occupying different roles. We next discuss these differences.

The Differential Effects of GenAI for Generalists and Specialists

Employees who face higher translation and coordination challenges or whose knowledge can be more powerfully catalyzed by grounded GenAI should display distinct interaction patterns after adoption. We draw on prior research to argue that generalists face fewer coordination challenges when collaborating, and that their knowledge is less likely to become attractive to others post GenAI adoption, than specialists. Unlike specialists, who cultivate deep expertise within a narrow domain (Becker 1962), generalists develop sufficient, shallower competence across multiple domains (Teodoridis 2018, Raveendran et al. 2022). As a result, generalists are more likely to have knowledge overlaps with those they interact with, making collaboration with others in general easier for them, than it is for specialists. The gains from being able to allocate freed-up time to collaboration should also be relatively modest for generalists. For example, in

the context of open-source software development, recent research has found that those who were initially more constrained in collaboration (i.e. the specialists) derived greater network benefits from GenAI (Hoffmann et al. 2025).

Further, the increase in value of a specialist’s knowledge to others after GenAI adoption should be greater than the corresponding increase in the value of a generalists knowledge to others. Specialists develop expertise through years of study and practice in a narrow domain, which entails knowledge that may not be easy to share, but is more valuable when shared. GenAI should therefore create a stronger catalyzing effect on the knowledge of specialists. We therefore expect a net effect of greater in-degree centrality for specialists in the knowledge network, and for both in- and out-degree centrality for specialists in the collaboration network.

Hypothesis 2a. Following the adoption of grounded GenAI, specialists increase both their in-and out- degree centrality in the collaboration network more than generalists.

Hypothesis 2b. Following the adoption of grounded GenAI, specialists increase their in-degree centrality in knowledge networks more than generalists.

Our hypotheses together capture the expected overall network effect of the adoption of grounded GenAI (Hypothesis 1) and the nuanced ways these effects diverge for generalist versus specialist roles (Hypotheses 2a and 2b). Our predictions extend the dominant, individual-centric narrative on GenAI by treating collaboration as an adjustable margin, one that GenAI plausibly shifts through automation and augmentation of context-specific work (Brynjolfsson et al. 2021, Brynjolfsson et al. 2023, Zheng et al. 2023, Gupta et al. 2024) and by linking those shifts to concrete network consequences.

Research Design

To test our theory, we conducted a randomized controlled trial with pre- and post-intervention assessments at a technology service provider company in Central Europe. The organizational context involved knowledge-intensive work and team collaboration, making it a suitable setting to examine the effects of grounded GenAI on output and network dynamics. Teams within this organization operated independently as each team in this organization focused on distinct products and services, which limited inter-team influence. This setup allowed a controlled implementation of grounded GenAI in some teams (treatment) but not in others (control). By using a randomized field experimental design with clear separation between treatment and control groups, we ensured that any differences observed post-intervention can be causally attributed to the implementation of grounded GenAI.

The field experiment was conducted within the normal working environment of the company. The GenAI tool was customized using RAG to generate grounded models building on GPT-4 and eventually GPT-4o as the foundational model. This produced an AI assistant connected to the organizational knowledge bases and communication data, enabling advanced knowledge retrieval and real-time support during work tasks. The AI assistant was capable of quickly searching and summarizing large volumes of internal documents, customer history from the customer relationship management (CRM) system, and prior communications (e.g., emails and meeting recordings) to provide relevant information on demand. For example, sales team members used the assistant to prepare for customer meetings by generating summaries of each customer's history, understand specific product or legal terms, technical product specifications, new regulations, past interactions, and relevant product and service information. Employees retrieved, created and provided relevant information through prompting the AI assistant. The system also assisted during live customer engagements in online meetings or calls by providing

answers or resources in real time. Finally, it helped organize follow-up tasks by summarizing meeting notes or extracting action items from communications, which was intended to improve the efficiency of managing projects.

Participants and Sample

The participants in the field experiment consisted of employees in three key roles: generalists - customer-facing sales personnel with direct client contact; specialists - technical product and service experts who supported the generalists; and managers - supervisors of these employees. While our primary interest was in comparing the behavior of generalists and specialists, the managers formed part of the organization under study and could not be excluded. In total, there were 333 employees in these roles. Employees working out their notice period or on temporary contracts were not included in the experiment, as they might not have stayed for the full duration of the experiment. Eventually, 318 employees responded to the survey (316 with a response in both, pre- and post-treatment assessment) and participated actively and voluntarily in the experiment.

The unit of randomization was the team level. The decision to randomize at the team level (cluster randomization) was made to avoid cross-contamination, as employees within the same team shared information frequently, while interaction across teams was limited due to the structure of the organization and incentive system. We randomly assigned each of the 42 teams in the organization to either the experimental condition (GenAI implementation) or the control condition (business as usual) using a computerized random draw to ensure impartial assignment. In total, 25 teams (197 employees) were randomly selected for the treatment group, and 17 teams (121 employees) for the control group.

Participants consisted of 217 generalists, 67 specialists and 34 managers. Within the experimental group, there were 137 generalists, 38 specialists and 22 managers; in the control group, there were 80 generalists, 29 specialists and 12 managers. Basic demographics (e.g., gender composition) were recorded: Overall, about 37% of participants were female. We verified that the randomization process produced comparable groups. There were no significant differences between the treatment and control teams in key baseline measures such as average output or network metrics before intervention (see Table 1 for covariate balance tests).

Intervention and Measurement

The intervention, in the form of a grounded GenAI assistant, was introduced to the experimental teams in September 2024. Training and awareness sessions were provided to teams to ensure they knew how to use the tool's features in their daily work processes. Training was delivered within existing teams related to their specific product and services domains. The teams in the control group continued with business as usual, meaning they did not have access to the AI tool and relied on existing resources and traditional methods for the same tasks (manual search in databases, personal expertise, contacting colleagues by email, etc.). This control condition provides a baseline to compare against the AI-assisted performance of the experimental group.

We carried out the data collection via online surveys administered at two points in time. Participants completed a baseline survey prior to the intervention and a nearly identical survey three months later capturing the same key variables. By using identical measures at two points, we were able to track changes over time within individuals and teams. The three-month interval was chosen to allow the technology to materialize effects in work behaviors and routines (Hollenbeck and Jamieson 2015, Costopulos Cappi and Accorinti 2017). Conducting the survey at multiple points in time enabled a longitudinal analysis of changes attributable to our

intervention, helping to capture how networks and employee performance evolved due to the implementation of GenAI.

Participation in the surveys was voluntary and confidential, but we achieved high response rates: 95.5% (318 out of 333) for the first survey and 95.2% (317 out of 333) for the second. These high response ratios were obtained due to regular executive communication, endorsement emails from senior leadership, and team managers encouraging participation and emphasizing that the insights from the research could benefit the teams and the organization. Management emphasized the importance of participating in the roll-out of grounded GenAI for improving personal productivity and overall competitive advantage. We collected answers from 318 individuals, but two respondents only had one answer each in the pre- or post-treatment survey. Therefore, the final sample size was 316 (all of whom provided both pre- and post-intervention responses). The high response rate reduces concern about nonresponse bias and strengthens the validity of the results (Baruch and Holtom 2008).

Variables

For our measures of collaboration networks, we assessed social interactions related to work: “With whom did you work closely during the last 3 months to achieve your objectives? (please name any relevant colleague, first name + last or email address).” For measuring the knowledge network, we asked: “To whom did you reach out during the last 3 months for expert advice regarding your core product / service? (please name any relevant colleague, first name + last or email address).” This involved employees identifying the relevant network actors for collaboration and knowledge networks, which included listing all relevant individuals involved in the respective networks (Tichy and Fombrun 1979, Wasserman 1994).

Degree, out-degree and in-degree for collaboration networks were calculated from the survey data following standard network analysis definitions to understand the network structures and respective change in structure post-intervention (Badar et al. 2015). *Degree* captures an individual's overall connectivity within a network and is defined as the number of direct ties an actor maintains relative to the total number of possible ties in that network. Higher degree centrality indicates greater activity and prominence in network interactions, reflecting more frequent involvement in collaboration or knowledge-sharing relationships (Hollenbeck and Jamieson 2015). *Out-degree* measures the number of outgoing ties from an individual . It shows how many connections an actor initiates, indicating their influence or activity within the network (Hollenbeck and Jamieson 2015, Baek and Bae 2019, Shrestha et al. 2025). *In-degree* measures the number of incoming ties to an actor. It indicates how many connections are directed towards a particular actor, reflecting their popularity or importance within the network (Tortoriello et al. 2012, Hollenbeck and Jamieson 2015, Baek and Bae 2019).

Ethical Considerations

We adhered strictly to data protection regulations and the company's privacy policies. Participation was voluntary at all times of the experiment. Before the initial survey, all eligible employees received an explanation of the purpose of the experiment. Employees had to opt in by agreeing to an informed consent statement embedded at the start of the survey. This statement assured them that their responses would be confidential, used only for research purposes, and reported in aggregate without identifying any individual. Participants were also informed of their right to withdraw from the study at any time without any consequences. To protect privacy, the surveys were confidential (we used coded IDs to match pre- and post-responses for analysis). As noted, performance data was self-reported rather than extracted directly, in order to respect

personal data protection guidelines. Additionally, when constructing the collaboration and knowledge networks, we treated lists of names with strict confidentiality. Network data was only used to compute metrics. The design of the field experiment was reviewed and approved by the company's compliance officer.

Analytical Approach

Our analytical approach was designed to leverage the experimental design and longitudinal data to identify the causal impact of grounded GenAI adoption on different outcomes. We tested for randomization. For each dependent variable, we computed difference scores by subtracting pre-treatment from post-treatment values. We then estimated a series of linear regressions using these difference scores as the outcome variables. The primary independent variable was a binary indicator for treatment assignment, which we interacted with a dummy variable for generalist status to assess heterogeneous treatment effects. Standard errors were clustered at the team level to account for intra-team correlation (Aguinis et al. 2013). Analyses were conducted at the individual level ($N = 316$), and we qualitatively replicated the findings using team-level aggregated data ($n = 42$).

Results

Descriptive statistics and zero-order correlations are shown in Tables 2a and 3. Hypothesis 1 states that the adoption of grounded GenAI increases connectivity, the number of ties, in both collaboration and knowledge networks. Our results provide strong support for this hypothesis. For collaboration networks, degree centrality increased by 7.77 in the treatment group compared to 1.12 in the control group ($M_{AI} - M_{control} = 6.65$; $p < .001$). Similar patterns are observed for in-degree and out-degree centrality, as shown in Table 2a and Figure 2. For knowledge networks, degree centrality increased by 5.21 in the treatment group compared to

0.84 in the control group ($M_{AI} - M_{control} = 4.37, p < .001$). In-degree and out-degree centrality exhibit similar patterns, as illustrated in Table 2a and Figure 2.

The results indicate that the adoption of the GenAI tool significantly enhances connectivity in both collaboration and knowledge networks. Individuals with access to the tool show substantially larger increases in total degree, in-degree, and out-degree centrality than those in the control group. The treatment effects reported in Table 4a further corroborate these findings and provide strong support for Hypothesis 1. Figure 1 shows the collaboration and knowledge sharing networks before and after GenAI adoption. As is evident, overall network connectivity increases, but particularly strongly among the treatment teams, consistent with the estimated positive treatment effects on both in and out-degree centrality.

Hypothesis 2a states that, following the adoption of grounded GenAI, specialists experience larger increases in collaboration network centrality (both in and out degree) than generalists. However, results indicate that adoption of GenAI is associated with substantial increases in centrality measures for both specialists and generalists. Degree centrality increased by 7.34 for specialists and by 8.71 for generalists, in-degree centrality increased by 3.26 for specialists and by 4.45 for generalists, and out-degree centrality increased by 4.08 for specialists and by 4.26 for generalists as highlighted in Table 2b and 2c. As shown in Table 4b, the interaction terms capturing the differential effects of GenAI adoption for generalists relative to specialists are not statistically significant for degree centrality ($b = -0.151; p > .10$), in-degree centrality ($b = 0.204; p > .10$), or out-degree centrality ($b = -0.506; p > .10$). Thus, both groups experienced increases in collaboration centrality and no differential effect by role was observed. Hence, Hypothesis 2a is not supported.

Hypothesis 2b posits that specialists' knowledge network in-degree centrality increases more than that of generalists following the adoption of grounded GenAI. Results for the knowledge network indicate specialists experienced an increase of 6.26 in their in-degree centrality whereas generalists only experienced an increase of 2.28. As shown in Table 4b, the interaction between GenAI adoption and generalist status for in-degree centrality was negative and significant ($b = -3.954$; $p < .05$), indicating that generalists experienced a smaller gain. Degree centrality increased by 9.76 for specialists and by 4.74 for generalists and out-degree centrality increased by 3.50 for specialists and by 2.46 for generalists. These results suggest that specialists became more important as distributors and sources of knowledge in the network after GenAI adoption than generalists, consistent with Hypothesis 2b.

Generalists and specialists are not randomly assigned categories in our setting and differ markedly in their pre-treatment levels on key outcome measures. As a result, the H2 results cannot be interpreted as causal subgroup effects in the same way as the overall treatment effect in Hypothesis 1 and are instead interpreted as correlational patterns describing how grounded GenAI adoption is associated with outcome changes across roles. To assess whether baseline comparability could be improved, we conducted a series of matching- and weighting-based robustness checks restricted to specialists and generalists, using pre-treatment outcome measures as balancing covariates. Across specifications, these diagnostics indicated very limited common support between the two role groups. Matching approaches substantially reduced sample size while leaving key baseline imbalances unresolved, and propensity-score-based approaches relied on very small effective samples or narrow overlap populations. Taken together, these checks reinforce the conclusion that causal comparisons between specialists and generalists are difficult

to support in this setting, and that the role-based results should be interpreted with appropriate caution.

Additional analysis

While our primary contribution concerns network structures, we also examine output as a secondary outcome, because network changes and individual task efficiencies can jointly translate into measurable work. Although generalists may not see as large an increase in connectivity, we might expect their output gains to exceed those of specialists (Fahrenkopf et al. 2020). Generalists' effectiveness comes from integrating diverse inputs and coordinating across boundaries, activities where GenAI's assistance can amplify productivity. By partially automating routine parts of both individual work and coordination, GenAI should allow generalists to focus more on high-level integration, leveraging their broad skill sets (Koehler et al. 2024). In other words, we expect the "self-automation" effects to translate into productivity gains to a greater extent for generalists than specialists.

In our study, we measured output using the number of projects managed, since GenAI tools can enhance this outcome by increasing employee productivity (Brynjolfsson et al. 2023, Dell'Acqua et al. 2023, Noy and Zhang 2023, Boussioux et al. 2024). We measured employee output by asking participants to report, based on their current CRM records, the *number of projects* they had worked on during the past three months ("On how many customer projects did you work / have you actively supported during the last 3 months based on your current CRM reporting?"). The number of projects was self-reported and not centrally extracted and mapped onto participants to respect data protection policies. As an additional subjective measure of work-related outcomes, we also included a measure of participants' *satisfaction* with their perceived knowledge access when requesting advice within their intra-organizational networks.

The results of our analysis show that generalists' output, measured by projects managed, was significantly higher after the implementation of grounded GenAI ($M_{AI,post} = 16.30$; $SD_{AI,post} = 4.63$) compared to the baseline output before the implementation ($M_{AI,pre} = 12.7$; $SD_{AI,pre} = 4.63$; $t = 13.17$; $p < .001$). The results for specialists show similar significant patterns after the implementation of grounded GenAI ($M_{AI,post} = 5.42$; $SD_{AI,post} = 3.58$) compared to the baseline output before the implementation ($M_{AI,pre} = 4.29$; $SD_{AI,pre} = 3.08$; $t = 6.39$; $p < .001$). As shown in Table 4b, the interaction term between GenAI adoption and generalist status for output was positive and significant ($b = 2.48$; $p < .01$), indicating that generalists experience a substantially higher absolute increase in output than specialists. Notably, even though the percentage increase in output for generalists (28.3%) is only marginally higher than that for specialists (26.4%), this is because of their different baselines. The interaction effect indicates that the absolute gain was larger for generalists. Overall, our findings indicate that generalists translated the adoption of grounded GenAI into greater output gains than specialists, with differences more pronounced in absolute terms than in relative terms.

We also observed a strong negative correlation between individuals' knowledge-network in-degree centrality and their output (the number of projects they worked on). This means that in our sample, employees handling many projects tended to be less central in the knowledge-sharing network, whereas those involved in fewer projects were more often regarded as knowledge sources.

Follow-up interviews

To provide additional insight into the mechanisms underlying these quantitative patterns, we conducted a small set of follow-up interviews with employees in the treatment group (seven generalists and six specialists). While the interview sample is limited and not intended for

systematic qualitative inference, participants' accounts are broadly consistent with the observed results. Generalists emphasized that the grounded GenAI assistant reduced time spent on routine information retrieval and follow-up tasks, allowing them to resolve simpler requests independently and to reallocate time toward coordinating more complex projects with colleagues (self-automation). Specialists, in contrast, highlighted that the tool enabled faster synthesis and higher-quality preparation of outputs, which increased the frequency with which others sought their advice (knowledge catalyst). These accounts offer illustrative support for the self-automation and knowledge-catalyst mechanisms discussed above, while remaining indicative given the small interview sample.

Discussion

Our field experiment provides causal evidence that integrating GenAI grounded in firm-specific knowledge into employees' workflows can significantly reshape collaboration patterns and improve output. First, confirming Hypothesis 1, we found that employees with access to the grounded GenAI assistant experienced substantial increases in their collaboration and knowledge-sharing network centrality relative to the control group. This result supports the idea that AI augmentation encourages more frequent collaboration by reducing coordination effort and/or making collaboration more valuable thereby rewiring the organization's social network for higher connectivity (Kilduff and Brass 2010). This supports our expectations that at the current state of AI deployment, the self-automation and knowledge catalyst effects of GenAI adoption are likely to be stronger in aggregate than modularization and knowledge substitution effects.

Second, Hypothesis 2a and 2b posited that generalists would gain less connectivity from GenAI adoption in collaboration and knowledge networks than specialists. Our results provide

only partial support for this hypothesis. Generalists and specialists exhibited significant and nearly equal increases in overall degree centrality after adopting grounded GenAI, indicating that connectivity gains accrued broadly across roles. However, specialists did realize greater gains in being sought out for knowledge. This pattern of results (no differential effect in overall collaboration ties, but a strong differential effect in knowledge centrality) suggests that GenAI’s “knowledge catalyst” effect benefited specialists’ expertise to a greater extent, though both specialists and generalists expanded their networks (i.e. our evidence does not suggest a weaker “knowledge substitution” effect for specialists). In our study, grounded GenAI’s networking benefits accrued to all users rather than being sharply differentiated by role, with the only notable role difference being the larger boost in knowledge centrality for specialists.

The interviews help contextualize these patterns and shed light on how employees translated grounded GenAI adoption into changes in work and interaction. Generalists described using the AI assistant to handle routine knowledge requests more autonomously, which reduced reliance on specialists for “easy” questions and freed capacity for managing a larger number of projects and coordinating across boundaries. As one generalist noted, “I can deliver some things straight away on my own now, and that gives me time to involve others when projects are more complex.” Specialists, in turn, emphasized that faster retrieval and synthesis allowed them to invest more attention in each request, increasing the perceived quality of their contributions and their visibility as experts. As one specialist put it, “because the answers are better prepared, people come back more often to ask me for advice.” Although these qualitative insights are based on a small and non-representative interview sample, they are consistent with the quantitative evidence and help illustrate how grounded GenAI can simultaneously expand collaboration and amplify role-specific advantages.

In additional analysis, we also found that generalists using the GenAI assistant completed more projects over the study period than their control-group counterparts, and these gains were higher than those observed for specialists. This role-based divergence in output aligns with theoretical expectation that generalists, whose work inherently involves integrating diverse inputs and coordinating across boundaries, can leverage GenAI to achieve larger productivity gains (Raisch and Krakowski 2021). These results resonate with recent research showing that AI tools tend to boost performance especially for those in roles or at skill levels where integrating knowledge from multiple sources is key (Hui et al. 2024, Jia et al. 2024). Notably, specialists also saw a significant uptick in output with grounded GenAI (supporting a broad augmentation effect), but their gains were comparatively smaller.

Taken together, these findings draw a nuanced picture. GenAI adoption simultaneously “rewires” the intra-firm network, making employees more connected in both work and knowledge relationships, and boosts productivity, with some differentiation across roles in output but not in overall networking gains. In other words, the introduction of GenAI in our setting enhanced connectivity and knowledge flow among all employees, while allowing generalists to translate these benefits into greater performance improvements than specialists, and specialists into greater influence within the network as sources of knowledge than generalists. This provides causal evidence that GenAI tools do not merely alter individual productivity; they can reshape human-human interaction patterns in organizations and do so in ways that increase users’ centrality in social networks (both as knowledge sources and collaborators).

Theoretical Implications

Our study makes several contributions to management theory and the growing literature on human-AI collaboration. First, we advance the conversation beyond the individual level by demonstrating how AI tools can induce systemic, network-level changes inside organizations. Prior research on generative AI in workplaces has largely focused on individual productivity gains and task performance, such as how much faster or better a person can write code or content with AI support, often within the framework of the automation-augmentation debate (Noy and Zhang 2023, Boussieux et al. 2024, Hui et al. 2024). These studies show that AI can augment human output under certain conditions but tend to treat co-workers and the structure of collaboration as static background (Kilduff and Brass 2010).

By contrast, our findings empirically validate the notion that AI's impact extends into the social fabric of the organization. In line with calls for a more network-centric or system-level view of technology's role in organizations (Zammuto et al. 2007, Anthony et al. 2023), we provide causal evidence that introducing a “digital colleague” (here, a grounded GenAI assistant) alters who talks to whom, how much, and with what outcomes. To our knowledge, this is among the first field experiments to show that an AI tool can reconfigure collaboration patterns in a measurable way, thereby bridging the gap between micro-level AI effects and macro-level organizational change (Bankins et al. 2024). This extends classic notions from organization theory that new technologies can change the structure of work and interaction (Zammuto et al. 2007), bringing that idea into the era of GenAI by documenting actual shifts in network structure following AI adoption.

Second, our results inform theory on the complementarity between human and AI capabilities, contributing to the automation-augmentation discourse (Jarrahi 2019, Raisch and Krakowski 2021). We observed a predominantly augmentation-oriented effect in our context:

The AI did not replace human interaction or work; instead, it amplified them. Users of the GenAI became more central in knowledge networks and increased their output, reflecting a scenario where automation of certain tasks (information search, initial drafting, and routine coordination) freed up capacity for more collaboration and production, a virtuous cycle of reinforcement between solo work and joint work.

This outcome provides empirical support for theoretical arguments that AI can simultaneously automate some components of a job while augmenting the role of humans in coordination and integrative tasks (Tang et al. 2022, Vaccaro et al. 2024). In particular, our finding that generalists saw especially large performance boosts with AI is consistent with the idea of human-AI complementarity: When an enabling technology raises the returns on both individual and collaborative effort, those who rely on blending those efforts (generalists) stand to gain the most (Milgrom and Roberts 1990, Kretschmer and Puranam 2008).

Third, our work provides new insights into the knowledge-sharing and social capital implications of GenAI. By boosting network connectivity and the flow of information between employees, grounded GenAI appears to function as a knowledge catalyst in the organization. The AI system effectively reduces the cost of seeking and providing knowledge, thereby strengthening the organization's internal knowledge circulation (Argote et al. 2022). In our study, employees with the GenAI assistant had more colleagues turning to them for expertise and, in turn, they reached out to more colleagues compared to the control group.

This can be interpreted through a knowledge brokerage lens: Those “augmented” by GenAI became more pivotal knowledge brokers in their networks (Waardenburg et al. 2022). In network terms, users of the AI increased both their in-degree and out-degree centrality, which can confer advantages like access to diverse information, influence, and opportunities for

innovation (Freeman 2002). Notably, this structural change in networks was accompanied by higher self-reported productivity and satisfaction with knowledge access, suggesting a positive link between augmented social capital and individual work-related attitudes and behaviors. The theoretical implication is that AI adoption can be a driver of organizational social dynamics, not just a technical aid for task efficiency. Our findings thus demonstrate that an AI tool can alter the distribution of social capital in an organization.

We echo recent voices in the management literature (Anthony et al. 2023) in urging a more holistic “system” view of AI in organizations, where the unit of analysis is not only the task or the individual user, but the configuration of people and AI agents in a collective system of work. Our study provides concrete evidence for this view by showing how introducing AI can ripple through and affect collaborative structures and group-level outcomes, linking micro-interventions to macro consequences. In summary, our research positions AI not just as an exogenous technological input, but as an endogenous factor in organizational design, one that influences collaboration networks, knowledge flows, and the balance of work across different human roles. This contributes to a deeper theoretical understanding of how augmentation plays out across various contexts and roles and underscores the importance of considering organizational networks when studying the impact of AI.

Practical Implications

For managers and organizations, these findings provide several actionable insights. Adopting a grounded GenAI system can be a powerful way to enhance internal collaboration and productivity, but it requires thoughtful implementation to fully capture the benefits.

When employees are armed with an AI that can quickly fetch information, draft communications, or provide answers, they appear more (not less) inclined to reach out to

colleagues as well as share knowledge, fostering a more networked and knowledge-rich culture.. Managers should therefore view GenAI not just as an efficiency tool for individual tasks, but as a form of collaboration infrastructure that can reshape team dynamics and information flows (Tarafdar et al. 2019). By lowering the effort required to seek help or expertise, the AI encouraged people to leverage their peers more often, a reminder that technology and human teamwork go hand in hand (Wilson and Daugherty 2018). Leaders may need to train employees not only in using the AI's features, but also in new collaboration practices that capitalize on the AI's presence.

Another implication relates to role management and ensuring equitable benefits of AI across different employee groups. We found that both customer-facing staff (generalists) and technical experts (specialists) benefited in their network connectivity from the GenAI deployment, even though performance gains were larger for generalists, and influence gains (centrality in in-degree terms) were larger for specialists. This suggests that organizations deploying AI tools must consider possible exacerbation of inequalities between roles. On the positive side, everyone can become more connected and visible in the network, as evidenced by the parallel increases in ties for both types of employees. From an inclusion perspective, this is encouraging, GenAI, at least in our case, did not only help an elite subset of employees but broadly enhanced connectivity.

However, the larger productivity gains observed for generalists suggest companies might realize especially high returns by focusing AI support on roles that integrate information across domains (e.g., project managers, coordinators, or product owners). The greater influence gains of specialists could alter the political dynamics of the organization. Further, increased connectivity is in general positive for knowledge flow, but it can potentially lead to information overload or

coordination fatigue if not well managed. In summary, grounded GenAI can be a tool for organizational transformation, improving how people connect and perform. Our empirical evidence of augmentation and network enrichment, as opposed to automation-led isolation, supports the view that AI can enhance human work (Puranam, 2024).

Limitations and Future Research

Despite the strengths of our randomized field experiment, several limitations must be acknowledged, which also point to avenues for future research. First, our study was conducted in a single organization. While this allowed for a controlled examination of AI's impact in one context, it does limit the generalizability of the findings. The company's organizational culture, industry (tech services), and geographic region (Central Europe) may have influenced how grounded GenAI affected collaboration and performance. Future studies should examine similar interventions in different settings, across multiple industries, organizational cultures, and regions, to see if comparable network and productivity effects occur.

Second, it would be valuable to explore longer-term dynamics. Our post-intervention window was three months, which captured the initial effects of GenAI adoption, but longer-term studies could reveal whether these network changes persist, are amplified, or even reverse over time as people and technology co-evolve.

Third, our measures of collaboration and knowledge ties relied primarily on self-reported survey data. This approach is common in organizational network research (Tichy and Fombrun 1979, Wasserman and Faust 1994) and was appropriate given our need to capture perceived working and advice relationships. However, survey-based network data can be susceptible to recall bias or respondent perception errors. Participants might forget certain interactions or might over/under-report ties based on recent experiences or subjective impressions. Importantly, our in-

degree measures are based on reports from others in the network, providing a more objective perspective on individuals' prominence as collaborators or knowledge sources.

Future research can strengthen this area by complementing surveys with digital trace data (e.g., communication metadata, email logs, and collaboration platform records) to map communication networks more objectively and at scale (Hollenbeck and Jamieson 2015, Polzer 2022, Cardon et al. 2023). Such data could help triangulate the survey results and distinguish between informal emergent ties and formal work interactions. Moreover, incorporating qualitative data (e.g., interviews or focus groups) would enrich our understanding of how employees decide to use the GenAI assistant and how it influences their work habits. Hearing directly from participants about their experiences with the AI why they chose to consult it for certain tasks or how it affected their willingness to reach out to coworkers could provide insight into the mechanisms underlying the quantitative patterns we observed.

Fourth, we acknowledge the challenges in interpreting the differences between generalist and specialist employees in our results, since these roles are not randomly assigned within the organization. Although the adoption of grounded GenAI was randomized at the team level, generalists and specialists differ systematically in their pre-treatment levels on key outcome measures, including network position and output. As a consequence, differences in how outcomes change across roles cannot be interpreted as causal estimates of heterogeneous treatment effects. Rather, the role-based results for Hypotheses 2a and 2b should be understood as correlational patterns, describing how the association between GenAI adoption and changes in network position and output varies across roles as they exist in the organization. Fifth, while we draw on a small set of follow-up interviews to illustrate potential mechanisms, these qualitative

insights are limited in scope and should be interpreted as indicative rather than as systematic evidence.

Conclusion

This research demonstrates that introducing a GenAI assistant that is contextually grounded in organization specific information can fundamentally alter how employees collaborate and perform. Through a randomized field experiment, we showed that grounded GenAI not only boosts individual productivity but also reconfigures the social networks within the firm, increasing connectivity and knowledge sharing among employees. These findings shift the lens from viewing AI purely as a tool to increase efficiency to seeing it as an agent of organizational change, one that can amplify human capabilities and reshape the collaborative landscape. GenAI can allow those in generalist roles to especially accelerate their performance, while allowing for those in specialized roles to increase their influence as sources of knowledge, illustrating how AI can have differential impacts on output even as it broadly enhances collaboration.

References

- Agrawal A, Gans J, Goldfarb A (2022) ChatGPT and how AI disrupts industries. *Harv. Bus. Rev.*, 12, 1-6.
- Aguinis H, Gottfredson RK, Culpepper SA (2013) Best-practice recommendations for estimating cross-level interaction effects using multilevel modeling. *J. Manag.* 39(6):1490–1528.
- Allen R, Choudhury P (2022) Algorithm-augmented work and domain experience: The countervailing forces of ability and aversion. *Organ. Sci.* 33(1):149–169.
- Anthony C, Bechky BA, Fayard AL (2023) “Collaborating” with AI: Taking a system view to explore the future of work. *Organ. Sci.* 34(5):1672–1694.
- Argote L, Guo J, Park SS, Hahl O (2022) The mechanisms and components of knowledge transfer: The virtual special issue on knowledge transfer within organizations. *Organ. Sci.* 33(3):1232–1249.
- Badar K, Hite JM, Ashraf N (2015) Knowledge network centrality, formal rank and research performance: Evidence for curvilinear and interaction effects. *Scientometrics* 105:1553–1576.
- Baek SI, Bae SH (2019) The effect of social network centrality on knowledge sharing. *Journal of Service Science Research* 11:183–202.
- Baldwin CY, Clark KB (2000) *Design Rules, Volume 1: The Power of Modularity*. (MIT Press).
- Bankins S, Ocampo AC, Marrone M, Restubog SLD, Woo SE (2024) A multilevel review of artificial intelligence in organizations: Implications for organizational behavior research and practice. *J. Organ. Behav.* 45(2):159–182.
- Baruch Y, Holtom BC (2008). Survey response rate levels and trends in organizational research. *Hum. Relat.* 61(8):1139–1160.

- Becker GS (1962) Investment in human capital: A theoretical analysis. *J. Political Econ.* 70(5, Part 2):9–49.
- Boussioux L, Lane JN, Zhang M, Jacimovic V, Lakhani KR (2024) The crowdless future? Generative AI and creative problem-solving. *Organ. Sci.* 35(5):1589–1607.
- Brynjolfsson E, Li D, Raymond LR (2023) Generative AI at work (No. w31161). Working paper, *The Quarterly Journal of Economics.* 140(2), 889-942.
- Brynjolfsson E, Rock D, Syverson C (2021) The productivity J-curve: How intangibles complement general purpose technologies. *AEJ: Macroeconomics* 13(1):333–372.
- Cardon PW, Ma H, Fleischmann C (2023) Recorded business meetings and AI algorithmic tools: Negotiating privacy concerns, psychological safety, and control. *Int. J. Bus. Commun.* 60(4):1095–1122.
- Chen Z, Chan J (2024) Large language model in creative work: The role of collaboration modality and user expertise. *Manag. Sci.* 70(12):9101–9117.
- Choudhary V, Marchetti A, Shrestha YR, Puranam P (2023) Human-AI ensembles: When can they work? *J. Manag.* 01492063231194968.
- Costopulos Cappi M, Accorinti B (2017) Formal structure and informal networks in organizations. An application of the social network analysis to a real-world case.
- Dell’Acqua F, McFowland E, Mollick ER, Lifshitz-Assaf H, Kellogg K, Rajendran S, Kraymer L, Candelon F, Lakhani K (2023) Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. Working Paper No. 24-013, Harvard Business School Technology & Operations Management Unit.

- Dietvorst, BJ., Simmons JP, Massey C (2015) Algorithm aversion: people erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114.
- Fahrenkopf E, Guo J, Argote L (2020) Personnel mobility and organizational performance: The effects of specialist vs. generalist experience and organizational work structure. *Organ. Sci.* 31(6):1601–1620.
- Freeman LC (2002) Centrality in social networks: Conceptual clarification. *Soc. Netw.* 1:238–263.
- Gupta R, Nair K, Mishra M, Ibrahim B, Bhardwaj S (2024) Adoption and impacts of generative artificial intelligence: Theoretical underpinnings and research agenda. *Int. J. Inf. Manag. Data Insights* 4(1):100232.
- Hillebrand L, Raisch S, Schad J (2025) Managing with artificial intelligence: An integrative framework. *Acad. Manag. Ann.* 19(1):343–375.
- Hoffmann M, Boysel S, Nagle F, Peng S, Xu K (2024) Generative AI and the nature of work (No. 11479). CESifo working paper.
- Hollenbeck JR, Jamieson BB (2015) Human capital, social capital, and social network analysis: Implications for strategic human resource management. *Acad. Manag. Perspect.* 29(3):370–385.
- Hui X, Reshef O, Zhou L (2024) The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market. *Organ. Sci.* 35(6):1977–1989.
- Jarrahi MH (2019) In the age of the smart artificial intelligence: AI's dual capacities for automating and informing work. *Bus. Inf. Rev.* 36(4):178–187.

- Jia N, Luo X, Fang Z, Liao C (2024) When and how artificial intelligence augments employee creativity. *Acad. Manag. J.* 67(1):5–32.
- Kemp A (2024) Competitive advantage through artificial intelligence: Toward a theory of situated AI. *Acad. Manag. Rev.* 49(3):618–635.
- Kilduff M, Brass DJ (2010) Organizational social network research: Core ideas and key debates. *Acad. Manag. Ann.* 4(1):317–357.
- Koehler DP, Rausch A, Biemann T, Buechsenschuss R (2024). Expertise and specialization in organizations: A social network analysis. *Eur. J. Work Organ. Psychol.* 34(2):1–16.
<https://doi.org/10.1080/1359432X.2024.2387874>
- Kretschmer T, Puranam P (2008) Integration through incentives within differentiated organizations. *Organ. Sci.* 19(6):860–875.
- Maher ML, Ventura D, Magerko B (2023) The grounding problem: An approach to the integration of cognitive and generative models. *Proceedings of the AAAI Symposium Series*, 2(1):320–325. <https://doi.org/10.1609/aaais.v2i1.27695>
- Milgrom P, Roberts J (1990) The economics of modern manufacturing: Technology, strategy, and organization. *Am. Econ. Rev.* 80(3):511–528.
- Noy S, Zhang W (2023) Experimental evidence on the productivity effects of generative artificial intelligence. *Science* 381(6654):187–192.
- Polzer JT (2022) The rise of people analytics and the future of organizational research. *Res. Organ. Behav.* 42:100181.
- Puranam P (2018) *The Microstructure of Organizations* (Oxford University Press, Oxford, UK).
- Puranam P, Singh H, Chaudhuri S (2009) Integrating acquired capabilities: When structural integration is (un)necessary. *Organ. Sci.* 20(2):313–328.

- Raisch S, Krakowski S (2021) Artificial intelligence and management: The automation-augmentation paradox. *Acad. Manag. Rev.* 46:192–210.
- Raveendran M, Puranam P, Warglien M (2022) Division of labor through self-selection. *Organ. Sci.* 33(2):810–830.
- Salas E, Rosen MA (2010) Experts at work: Principles for developing expertise in organizations. Kozlowski SWJ, Salas E, eds. *Learning, Training, and Development in Organizations* (Routledge Taylor & Francis Group), 99–134.
- Sanchez R, Mahoney JT (1996) Modularity, flexibility, and knowledge management in product and organization design. *Strateg. Manag. J.* 17(S2):63–76.
- Shrestha YR, Buechsenschuss R, Tinguely PN (2025) Leveraging AI-enabled people analytics for designing organizations. Hoffman CH, ed. *Artificial Intelligence, Entrepreneurship and Risk: Reflections and Positions at the Crossroads between Philosophy and Management* (Springer Fachmedien Wiesbaden), 333–352.
- Tang PM, Koopman J, McClean ST, Zhang JH, Li CH, De Cremer D, Ng CTS (2022) When conscientious employees meet intelligent machines: An integrative approach inspired by complementarity theory and role theory. *Acad. Manag. J.* 65(3):1019–1054.
- Tarafdar M, Beath CM, Ross JW (2019) Using AI to enhance business operations. *MIT Sloan Manag. Rev.* 60(4).
- Teodoridis F (2018) Understanding team knowledge production: The interrelated roles of technology and expertise. *Manag. Sci.* 64(8):3625–3648.
- Tichy N, Fombrun C (1979) Network analysis in organizational settings. *Hum. Relat.* 32(11):923–965.

- Tortoriello M, Reagans R, McEvily B (2012) Bridging the knowledge gap: The influence of strong ties, network cohesion, and network range on the transfer of knowledge between organizational units. *Organ. Sci.* 23(4):1024–1039.
- Vaccaro M, Almaatouq A, Malone T (2024) When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nat. Hum. Behav.* 8:2293–2303.
- Waardenburg L, Huysman M, Sergeeva AV (2022) In the land of the blind, the one-eyed man is king: Knowledge brokerage in the age of learning algorithms. *Organ. Sci.* 33:59–82.
- Wasserman S, Faust K (1994) *Social Network Analysis: Methods and Applications*.
- Wilson HJ, Daugherty PR (2018) Collaborative intelligence: Humans and AI are joining forces. *Harv. Bus. Rev.* 96(4), 114-123.
- Zammuto RF, Griffith TL, Majchrzak A, Dougherty DJ, Faraj S (2007) Information technology and the changing fabric of organization. *Organ. Sci.* 18(5):749–762.
- Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, Stoica I (2023). Judging LLM-as-a-judge with MT-bench and chatbot arena. *Adv. Neural Inf. Process. Syst.* 36:46595–46623.

Table 1: Overview of covariate balance**Overview of covariate balance at the individual level (pre-treatment)**

Variable (pre-treatment)	$M_{\text{control,pre}}$	$M_{\text{AI,pre}}$	$M_{\text{AI,pre}} - M_{\text{control,pre}}$	<i>t</i> stat (Welch)	<i>df</i>	<i>p</i> value
Female	0.388	0.345	-0.043	-0.773	248.8	.440
Generalist	0.661	0.695	0.034	0.631	248.2	.528
Team size	11.975	13.802	1.827	1.842	264.5	.067
Satisfaction	3.314	3.244	-0.070	-0.399	251.8	.690
Projects	8.901	9.934	1.033	1.568	263.9	.118
Know degree	3.793	4.178	0.384	1.070	290.1	.285
KnowIn	1.909	2.076	0.167	0.464	285.0	.643
KnowOut	1.884	2.102	0.217	1.513	244.1	.131
Col degree	9.471	10.655	1.184	1.687	293.9	.093
ColIn	4.752	5.289	0.537	1.077	285.5	.283
ColOut	4.719	5.365	0.646	1.759	273.0	.080

Overview of covariate balance at the team level (pre-treatment)

Variable (pre-treatment)	$M_{\text{control,pre}}$	$M_{\text{AI,pre}}$	$M_{\text{AI,pre}} - M_{\text{control,pre}}$	<i>t</i> stat (Welch)	<i>df</i>	<i>p</i> value
Female	0.408	0.364	-0.044	-0.572	32.0	.571
Generalist	0.389	0.480	0.091	0.584	37.1	.563
Team size	7.944	8.360	0.416	0.173	35.9	.864
Satisfaction	3.660	3.632	-0.028	-0.089	36.6	.930
Projects	6.764	7.916	1.153	0.788	39.8	.435
Know degree	4.666	5.377	0.711	0.854	40.4	.398
KnowIn	3.127	3.421	0.294	0.309	40.9	.759
KnowOut	1.539	1.956	0.417	1.556	40.9	.127
Col degree	8.674	9.997	1.323	1.290	39.4	.205
ColIn	4.460	5.278	0.818	1.193	39.5	.240
ColOut	4.214	4.719	0.505	0.822	40.3	.416

Notes: At individual level: $N = 318$ ($n_{\text{control}} = 121$; $n_{\text{AI}} = 197$), assessment answers of all respondents pre-treatment. All variables reported in this table are measured prior to treatment; at team level: $n_{\text{control}} = 17$; $n_{\text{AI}} = 25$); *t* stats and (approximated) degrees of freedom from Welch *t* tests.

Table 2: Descriptive statistics**Table 2a: Pre- and post-statistics (all participants)**

	Pre-treatment		Post-treatment	
Variable	$M_{\text{control.pre}}$ ($SD_{\text{control.pre}}$)	$M_{\text{AI.pre}}$ ($SD_{\text{AI.pre}}$)	$M_{\text{control.post}}$ ($SD_{\text{control.post}}$)	$M_{\text{AI.post}}$ ($SD_{\text{AI.post}}$)
Knowledge networks				
Degree	3.824 (2.874)	4.178 (3.473)	4.664 (2.832)	9.391 (9.192)
In-degree	1.941 (2.92)	2.076 (3.425)	2.261 (2.883)	4.711 (8.212)
Out-degree	1.882 (1.276)	2.102 (1.203)	2.403 (1.361)	4.680 (2.218)
Collaboration networks				
Degree	9.571 (5.518)	10.655 (6.875)	10.689 (5.499)	18.426 (11.985)
In-degree	4.815 (4.029)	5.289 (4.759)	5.311 (3.933)	9.178 (9)
Out-degree	4.756 (3.064)	5.365 (3.372)	5.378 (2.98)	9.249 (4.927)
Satisfaction	3.303 (1.537)	3.244 (1.515)	3.336 (1.445)	4.873 (1.473)
Projects	8.731 (5.478)	9.934 (5.881)	8.387 (5.753)	12.706 (6.9)

Table 2b: Pre- and post-statistics (Generalists)

Generalists	Pre-treatment		Post-treatment	
Variable	$M_{\text{control.pre}}$ ($SD_{\text{control.pre}}$)	$M_{\text{AI.pre}}$ ($SD_{\text{AI.pre}}$)	$M_{\text{control.post}}$ ($SD_{\text{control.post}}$)	$M_{\text{AI.post}}$ ($SD_{\text{AI.post}}$)
Knowledge networks				
Degree	2.692 (2.217)	2.606 (1.313)	3.462 (2.322)	7.350 (7.992)
In-Degree	0.500 (1.884)	0.299 (0.869)	0.821 (1.884)	2.584 (6.975)
Out-Degree	2.192 (1.162)	2.307 (1.054)	2.641 (1.45)	4.766 (2.22)
Collaboration networks				
Degree	9.526 (5.839)	9.701 (5.745)	11.090 (5.636)	18.409 (12.22)
In-Degree	4.474 (4.474)	4.336 (4.253)	5.218 (4.25)	8.788 (9.529)
Out-Degree	5.051 (2.984)	5.365 (2.708)	5.872 (2.88)	9.620 (4.614)
Satisfaction	2.872 (1.506)	2.912 (1.447)	2.885 (1.328)	4.839 (1.501)
Projects	11.577 (4.534)	12.701 (4.629)	11.218 (5.018)	16.299 (4.632)

Table 2c: Pre- and post-statistics (Specialists)

Specialists	Pre-treatment		Post-treatment	
Variable	$M_{\text{control.pre}}$ ($SD_{\text{control.pre}}$)	$M_{\text{AI.pre}}$ ($SD_{\text{AI.pre}}$)	$M_{\text{control.post}}$ ($SD_{\text{control.post}}$)	$M_{\text{AI.post}}$ ($SD_{\text{AI.post}}$)
Knowledge networks				
Degree	5.759 (2.415)	7.789 (3.441)	6.862 (1.995)	17.553 (10.757)
In-Degree	4.586 (2.368)	6.211 (3.239)	4.931 (2.086)	12.474 (9.599)
Out-Degree	1.172 (0.71)	1.579 (1.13)	1.931 (0.883)	5.079 (2.097)
Collaboration networks				
Degree	7.690 (2.606)	9.105 (5.598)	7.586 (2.542)	16.447 (10.724)
In-Degree	4.276 (1.925)	5.132 (3.55)	4.034 (1.86)	8.395 (6.033)
Out-Degree	3.414 (1.88)	3.974 (3.275)	3.552 (1.862)	8.053 (5.844)
Satisfaction	4.034 (1.295)	4.132 (1.318)	4.069 (1.251)	5.500 (1.033)
Projects	3.828 (1.733)	4.289 (3.083)	3.483 (2.08)	5.421 (3.583)

Table 3: Zero-order correlations

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 KN - Degree (pre)															
2 KN - In-degree (pre)	0.92***														
3 KN - Out-degree (pre)	0.20***	-0.17**													
4 KN - Degree (post)	0.44***	0.40***	0.09												
5 KN - In-degree (post)	0.44***	0.47***	-0.07	0.96***											
6 KN - Out-degree (post)	0.18***	-0.02	0.54***	0.55***	0.30***										
7 Col - Degree (pre)	0.27***	0.23***	0.10	0.13*	0.09	0.15**									
8 Col - In-degree (pre)	0.29***	0.28***	0.02	0.16**	0.15**	0.09	0.87***								
9 Col - Out-degree (pre)	0.13*	0.07	0.16**	0.03	-0.01	0.17**	0.75***	0.34***							
10 Col - Degree (post)	0.11*	0.07	0.11*	0.22***	0.15**	0.28***	0.51***	0.43***	0.41***						
11 Col - In-degree (post)	0.12*	0.10	0.05	0.19***	0.16**	0.18***	0.45***	0.50***	0.19***	0.92***					
12 Col - Out-degree (post)	0.04	-0.01	0.16**	0.17**	0.09	0.33***	0.42***	0.15**	0.61***	0.76***	0.45***				
13 Satisfaction (pre)	0.18***	0.22***	-0.08	0.12*	0.14**	-0.03	0.05	0.07	0.00	0.02	0.05	-0.01			
14 Satisfaction (post)	0.14**	0.12*	0.04	0.21***	0.16**	0.23***	0.02	0.01	0.03	0.19***	0.14**	0.19***	0.71***		
15 Projects (pre)	-0.43***	-0.52***	0.20***	-0.24***	-0.30***	0.07	-0.12*	-0.21***	0.04	0.07	0.00	0.17**	-0.25***	-0.05	
16 Projects (post)	-0.43***	-0.52***	0.21***	-0.16**	-0.24***	0.18***	-0.12*	-0.20***	0.05	0.12*	0.02	0.24***	-0.30***	0.02	0.90***

* $p < .05$, ** $p < .01$, *** $p < .001$

Pre = Pre-implementation of grounded GenAI, Post = Post implementation of grounded GenAI

KN = Knowledge networks, Col = Collaboration networks

Table 4: Regression outputs: Treatment effects on outcome differences

4a Treatment effects on outcome differences – Individual level treatment effects (standard errors clustered at team level)

	Knowledge networks			Collaboration networks			Δ Satisfaction	Δ Projects
	Δ Degree	Δ In-degree	Δ Out-degree	Δ Degree	Δ In-degree	Δ Out-degree		
AI	4.371***	2.315***	2.058***	6.650***	3.393***	3.261***	1.596***	3.116***
SE (AI)	0.841	0.642	0.273	0.786	0.430	0.441	0.204	0.456
r^2	0.092	0.035	0.283	0.122	0.060	0.183	0.423	0.257
F stat	27.01	13.01	56.71	71.63	62.270	54.77	61.11	46.75
Prob(F)	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000

Notes: $N = 316$, # clusters = 42, $df = 314$

4b Treatment effects on outcome differences between generalists and specialists (standard errors clustered at team level)

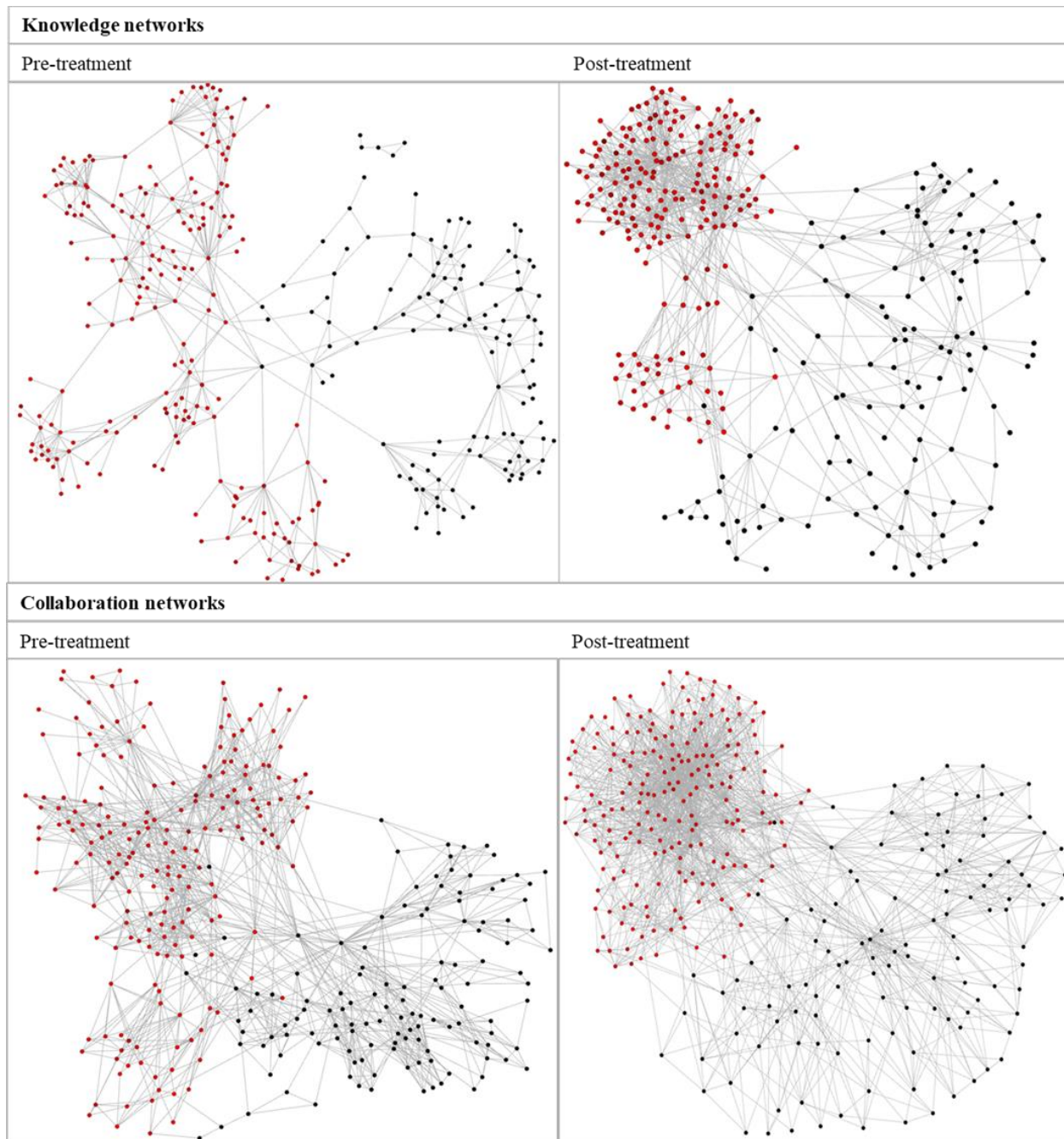
	Knowledge networks			Collaboration networks			Δ Satisfaction	Δ Projects
	Δ Degree	Δ In-degree	Δ Out-degree	Δ Degree	Δ In-degree	Δ Out-degree		
AI	8.659***	5.918***	2.741***	7.445***	3.505***	3.941***	1.334***	1.476***
SE (AI)	1.718	1.454	0.472	0.545	0.336	0.468	0.319	0.341
Generalist	-0.334	-0.024	-0.310	1.668***	0.985***	0.683***	-0.022	-0.014
SE (generalist)	0.452	0.281	0.231	0.261	0.210	0.144	0.178	0.314
AI x generalist	-4.684	-3.954***	-0.730	-0.302	0.204	-0.506	0.580*	2.481***
SE (AI x generalist)	1.862	1.521	0.580	0.881	0.534	0.644	0.329	0.541
r^2	0.161	0.092	0.328	0.147	0.075	0.213	0.546	0.362
F stat	18.697	12.021	23.566	129.068	73.127	53.349	193.750	38.865
Prob(F)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Notes: $n = 282$ (Managers are excluded from this analysis, which accounts for the smaller sample size reported in this table.), # clusters = 41, $df = 278$

OLS on outcome differences (post-pre). Standard errors clustered at the team level. Significance:

* $p < .05$, ** $p < .01$, *** $p < .001$. Coefficients reported with clustered SEs in parentheses.

Figure 1: Knowledge and collaboration networks pre- and post-treatment



Notes: Each node represents an employee in the experiment; $N = 316$.

Red nodes = treatment group. Black nodes = control group. Edge = relationship between employees.

Figure 2: Overview of results at the individual and team level

